

Save New Examples for Testing

Project: Helpful Classroom Sorter

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Build a Human Classifier

Without opening the old lesson, explain one key idea from Build a Human Classifier. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Debug the Cause

Without opening the old lesson, explain one key idea from Debug the Cause. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Algorithms as Recipes

Without opening the old lesson, explain one key idea from Algorithms as Recipes. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Preserve the original result for Keeping final tests unseen. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Keeping final tests unseen. Proposed explanation: “A new file makes old records unseen”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Run a separate ribbon experiment

Before receiving any test labels, freeze RibbonA: whole cm lengths 1 through 12, long at least 6 cm and short otherwise. Save predictions for 5, 7, 9 and 11 cm. Ask your teacher to reveal the reserved labels, then record agreements, disagreements and a limited result statement. If you later choose threshold 8 after seeing these labels, explain how you must describe its result on the same records.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Eight cards are used to invent a rule. Four sealed cards are then opened; the team changes its rule after seeing errors. Can those four cards still be called an untouched final test? Plan what to do next.

6. Solve the given case

Cards A–F are used for practice. G and H remain sealed.

Which can provide a fresh test, and under what condition?

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

You teach a sorter using cards A, B and C. You keep D and E unseen. Which cards provide the fairer first test? What changes if you show D during teaching?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 15 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Create separate learn and test envelopes for the Classroom Sorter and label who may open each.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Eight cards are used to invent a rule. Four sealed cards are then opened; the team changes its rule after seeing errors. Can those four cards still be called an untouched final test? Plan what to do next.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Assign the evidence its role

For each event, state whether examples are being used for development or independent evaluation of a fixed model: comparing candidates on fitting labels; scoring a frozen model on previously reserved examples once; changing a threshold after reading a test score. Explain your last choice.

My reasoning and evidence

13. W2 Preserve the original result

TowerA uses threshold 5 on reserved heights 3, 5, 7 and 9 cm. The labels are short, short, tall and tall. Write its predictions, agreements and disagreement. Then state why TowerB with threshold 6, selected after seeing these labels, cannot claim four independent test successes on these same records.

My reasoning and evidence

14. W3 Detect a less visible leak

A team leaves reserved labels out of its fitting table but tries six rules and reads a test score for each. It chooses the highest-scoring rule. Explain how information still reached model selection. Would renaming or reordering the same test records repair the problem?

My reasoning and evidence

15. W4 Run a separate ribbon experiment

Before receiving any test labels, freeze RibbonA: whole cm lengths 1 through 12, long at least 6 cm and short otherwise. Save predictions for 5, 7, 9 and 11 cm. Ask your teacher to reveal the reserved labels, then record agreements, disagreements and a limited result statement. If you later choose threshold 8 after seeing these labels, explain how you must describe its result on the same records.

My reasoning and evidence

16. W5 Audit a proposed final set

A class reserves ten copies of one easy development example in a new file. The model agrees with all ten labels. Explain two reasons this gives weak evidence about new cases, and propose a better final evaluation plan without promising perfect future performance.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Save New Examples for Testing

A fictional envelope exercise evaluates a fixed pencil-length sorter.

Freeze PencilP: long at least 6 cm; short below 6.

Reserved valid lengths are 3,5,6,9 cm. Teacher envelope labels, revealed only after predictions, are short,long,long,long.

Frozen predictions versus reserved labels

	Length cm	PencilP	Reference	Result
Test 1	3	short	short	agree
Test 2	5	short	long	disagree
Test 3	6	long	long	agree
Test 4	9	long	long	agree

Read a cell using both its row and column labels.

Save New Examples for Testing

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Freeze

Write PencilP and its boundary before opening the label envelope.



2. Predict

For 3,5,6,9 cm predict short,short,long,long.



3. Reveal references

Envelope labels are short,long,long,long.



4. Compare

Three of four agree; the disagreement is at 5 cm.

CHANGE ONE THING

After feedback, change the boundary to 5.

Explain the changed result using the matching lesson.

Does 4/4 on the replay replace the original 3/4 result? Explain.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Maintain dated sorter versions and separate development examples from reserved project-evaluation cases.

19. Project portfolio milestone

Project: Helpful Classroom Sorter

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Maintain dated sorter versions and separate development examples from reserved project-evaluation cases.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional envelope exercise evaluates a fixed pencil-length sorter.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Three of four agree; the disagreement is at 5 cm.

1. Reveal references
2. Compare
3. Freeze
4. Predict

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Compare Three of four agree; the disagreement is at 5 cm. The prediction record must exist before the labels are seen. PencilP scores $3/4=75\%$ agreement on these four selected cases. Moving the boundary after inspecting the envelope makes this a development exercise; the same four cases can no longer serve as untouched final evaluation.

Practice 2: explain the deciding evidence

Changed-case practice: After feedback, change the boundary to 5. Which conclusion is supported by the supplied facts?

1. The revised rule matches all four known labels but needs a different final test.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The revised rule matches all four known labels but needs a different final test. These labels influenced the revision, so replaying them is evidence of repair rather than unseen performance.

Repair a missing building block

Review Build a Human Classifier (ai-1/14). Goal: Construct a classifier with a stated input format, selected rule and recorded output. Short practice: Trace 0 cm and explain why clarification is different from predicting short.. Return to this week and explain what you changed.

Review Debug the Cause (ai-1/12). Goal: Compare expected and observed traces to locate the first difference. Short practice: Propose input 0 and give both predicted outputs without inventing an observed result.. Return to this week and explain what you changed.

Review Algorithms as Recipes (ai-1/9). Goal: Trace an ordered algorithm using the current value after every instruction. Short practice: Use the arrows to explain where the two zero-start runs first differ.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
